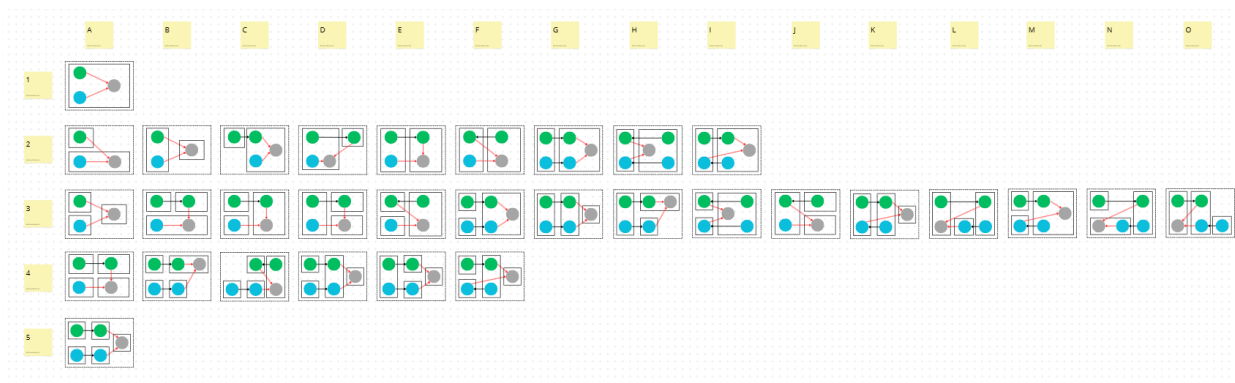


Confluence (CF)



A1- In the same file, a variable change in Left and a different variable change in Right cause interference in a third variable used as a basis.

```
Main.java x
1 public class Main { no usages
2     private static int VALOR_FINAL = 10; // right altera de 9 para 10
3     private static int ADD = 2; // left altera de 1 para 2 1usage
4
5     public static void main(String[] args) {
6         calcularValorFinal();
7         System.out.println("VALOR_FINAL é " + VALOR_FINAL);
8     }
9
10    private static void calcularValorFinal() { 1usage
11        // left e right queriam 11, merge resultou em 12
12        VALOR_FINAL = VALOR_FINAL + ADD;
13    }
14 }
```

Left increases the value of the variable 'ADD' by 1 on line 3. Right increases the value of the variable 'VALOR_FINAL' by 1 on line 2. Both expected the result of 'VALOR_FINAL' on line 12 to be 11, but the result in the merge was 12. This resulted in a CF conflict between lines 2 and 3 and line 11.

A2 - Modifying a variable using the Left method in one file. Modifying a variable using the Right method in a second file. In this second file, there is a base variable that uses the values modified by the Left and Right methods.

```

Main.java x
1  public class Main { no usages
2      private static int RESULTADO_FINAL = 0; 2 usages
3
4      public static void main(String[] args) {
5          Config.Y++; // Right incrementa o valor de Y
6          calcular();
7          System.out.println("RESULTADO_FINAL é " + RESULTADO_FINAL);
8      }
9
10     private static void calcular() { 1 usage
11         RESULTADO_FINAL = Config.X + Config.Y;
12     }
13 }

```

```

Config.java x
1  public class Config {
2      public static int X = 20; // left altera x de 10 para 20
3      public static int Y = 2;
4  }

```

Left modifies the value of the variable 'X' on line 2 of 'Config.java'. Right uses a command to change the value of the variable 'Y' on line 5 of 'Main.java'. These changes interfere with the result of the variable 'RESULTADO_FINAL' used by Base on line 11 of 'Main.java'. This creates a CF conflict that extends from lines 2 of 'Config.java' and 5 of 'Main.java' to line 11 of 'Main.java'.

B2 - Modifying a variable using the Left method in one file. Modifying another variable using the Right method in the same file. In a second file, there is a base variable that uses the values modified by the Left and Right methods.

```
Config.java ×
1 public class Config { no usages
2     public static int X = 20; // left altera x de 10 para 20
3     public static int Y = 3; // right altera y de 2 para 3 n
4 }

Main.java ×
1 public class Main { no usages
2     private static int RESULTADO_FINAL = 0; 2 usages
3
4     public static void main(String[] args) {
5         calcular();
6         System.out.println("RESULTADO_FINAL é " + RESULTADO_FINAL);
7     }
8
9     private static void calcular() { 1 usage
10         RESULTADO_FINAL = Config.X + Config.Y;
11     }
12 }
```

Left changes the value of the variable 'X' on line 2 of 'Config.java'. Right changes the value of the variable 'Y' on line 3 of 'Config.java'. These changes affect the result of the variable 'RESULTADO_FINAL' used by Base on line 10 of 'Main.java'. This creates a CF conflict that extends from lines 2 and 3 of 'Config.java' to line 10 of 'Main.java'.

C2 - Left adds a method call in one file, which changes the value of a variable in a second file. In that second file, Right changes the value of another variable. Still in the second file, these two changes generate a CF conflict for a variable used by Base that uses both values changed by Left and Right.

```

Main.java ×
1  public class Main { no usages
2      private static int RESULTADO_FINAL = 0; 1 usage
3
4      public static void main(String[] args) {
5          Config.xPlusTen(); // adição de left
6          Config.calcular();
7          System.out.println("RESULTADO_FINAL é " + RESULTADO_FINAL);
8      }
9  }

```

```

Config.java ×
1  public class Config { no usages
2      public static int X = 10; 2 usages
3      public static int Y = 2; 2 usages
4
5      private static void xPlusTen() {
6          X += 10; // adição de left
7      }
8
9      Y++; // adição de right
10
11     private static void calcular() {
12         RESULTADO_FINAL = X + Y;
13     }
14 }

```

In line 5 of 'Main.java', Left added a method call that changes the value of 'X' in line 6 of 'Config.java'. Right changes the value of 'Y' in line 9 of 'Config.java'. These changes interfere with the value of the variable 'RESULTADO_FINAL' used by Base in line 12 of 'Config.java'. This creates a CF conflict that extends from lines 2 of 'Config.java' and 5 of 'Main.java' to line 11 of 'Main.java'. This creates a CF conflict that originates in line 5 of 'Main.java' and goes from lines 6 and 9 of 'Config.java' to line 12 of 'Config.java'.

D2 - Left adds a method call in one file, which changes the value of a variable in a second file. In the first file, Right changes the value of another variable. Still in the first file, these two changes generate a CF conflict for a variable used by Base that uses both values changed by Left and Right..

```
☞ Main.java ×
1  public class Main { no usages
2      private static int RESULTADO_FINAL = 0; 2 usages
3
4      public static void main(String[] args) {
5          Config.xPlusTen(); // adição de left
6          Config.Y++; // adição de right
7          calcular();
8          System.out.println("RESULTADO_FINAL é " + RESULTADO_FINAL);
9      }
10
11     private static void calcular() { 1 usage
12         RESULTADO_FINAL = Config.X + Config.Y;
13     }
14 }
```

```
☞ Config.java ×
1  public class Config {
2      public static int X = 10;
3      public static int Y = 2;
4
5      private static void xPlusTen() {
6          X += 10; // adição de left
7      }
8  }
```

In line 5 of 'Main.java', Left added a method call that changes the value of 'X' in line 6 of 'Config.java'. Right changes the value of 'Y' in line 6 of 'Main.java'. These changes interfere with the value of the variable 'RESULTADO_FINAL' used by Base in line 12 of 'Main.java'. This creates a CF conflict that originates in line 5 of 'Main.java' and goes from lines 6 of 'Main.java' and 6 of 'Config.java' to line 12 of 'Main.java'.

E2 - Left adds a method call in one file, which changes the value of a variable in a second file. In the first file, Right changes the value of another variable. In the second file, these two changes generate a CF conflict for a variable used by Base that uses both values changed by Left and Right..

```
☞ Main.java ×
1  public class Main { no usages
2      public static void main(String[] args) {
3          Config.xPlusTen(); // adição de left
4          Config.Y++; // adição de right
5          Config.calcular();
6          System.out.println("RESULTADO_FINAL é " + Config.RESULTADO_FINAL);
7      }
8  }
```

```
☞ Config.java ×
1  public class Config { no usages
2      public static int X = 10; 2 usages
3      public static int Y = 2; 1 usage
4      private static int RESULTADO_FINAL = 0;
5
6      private static void xPlusTen() { no usages
7          X += 10; // adição de left
8      }
9
10     private static void calcular() { no usages
11         RESULTADO_FINAL = X + Y;
12     }
13 }
```

In line 3 of 'Main.java', Left added a method call that changes the value of 'X' in line 7 of 'Config.java'. Right changes the value of 'Y' in line 4 of 'Main.java'. These changes interfere with the value of the variable 'RESULTADO_FINAL' used by Base in line 11 of 'Config.java'. This creates a CF conflict that originates in line 3 of 'Main.java' and goes from lines 4 of 'Main.java' and 7 of 'Config.java' to line 11 of 'Config.java'.

F2 - Left adds a method call in one file, which changes the value of a variable in a second file. Also in the second file, Right changes the value of another variable. In the first file, these two changes generate a CF conflict for a variable used by Base that uses both values changed by Left and Right..

```
Main.java ×
1 public class Main { no usages
2     private static int RESULTADO_FINAL = 0; 2 usages
3
4     public static void main(String[] args) {
5         Config.xPlusTen(); // adição de left
6         calcular();
7         System.out.println("RESULTADO_FINAL é " + RESULTADO_FINAL);
8     }
9
10    private static void calcular() { 1 usage
11        RESULTADO_FINAL = Config.X + Config.Y;
12    }
13 }
```

```
Config.java ×
1 public class Config { no usages
2     public static int X = 10; 1 usage
3     public static int Y = 3; // alteração de right
4
5     private static void xPlusTen() { no usages
6         X += 10; // adição de left
7     }
8 }
```

In line 5 of 'Main.java', Left added a method call that changes the value of 'X' in line 6 of 'Config.java'. Right changes the value of 'Y' in line 3 of 'Config.java'. These changes interfere with the value of the variable 'RESULTADO_FINAL' used by Base in line 11 of 'Main.java'. This creates a CF conflict that originates in line 5 of 'Main.java' and goes from lines 3 and 6 of 'Config.java' to line 11 of 'Main.java'.

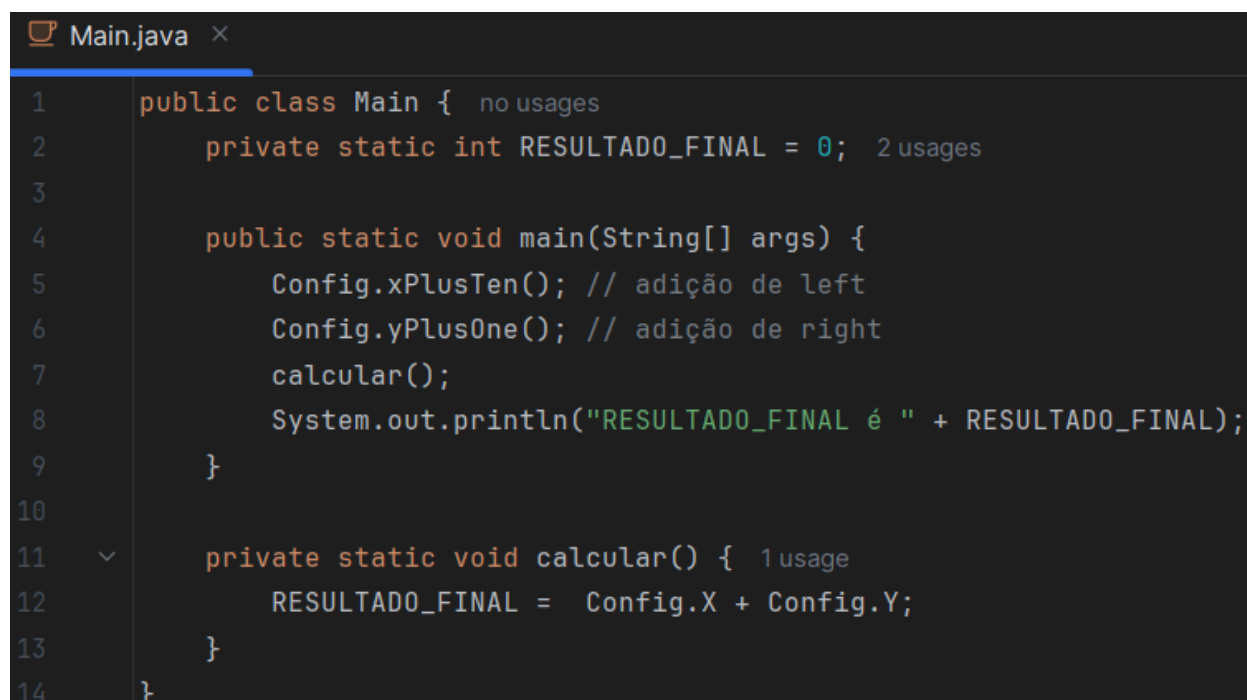
G2 - Left and Right add a method call in one file, which changes the value of different variables in a second file. In the second file, these two changes generate a CF conflict for a variable used by Base that uses both values changed by Left and Right..

```
Main.java x
1 public class Main { no usages
2     public static void main(String[] args) {
3         Config.xPlusTen(); // adição de left
4         Config.yPlusOne(); // adição de right
5         Config.calcular();
6         System.out.println("RESULTADO_FINAL é " + Config.RESULTADO_FINAL);
7     }
8 }
```

```
Config.java x
1 public class Config { no usages
2     public static int X = 10; 2 usages
3     public static int Y = 2; 2 usages
4     private static int RESULTADO_FINAL = 0;
5
6     private static void xPlusTen() { no usages
7         X += 10; // adição de left
8     }
9
10    private static void yPlusOne() { no usages
11        Y++; // adição de right
12    }
13
14    private static void calcular() { no usages
15        RESULTADO_FINAL = X + Y;
16    }
17 }
```

In line 3 of 'Main.java', Left added a method call that changes the value of 'X' in line 7 of 'Config.java'. In line 4 of 'Main.java', Right added a method call that changes the value of 'Y' in line 11 of 'Config.java'. These changes interfere with the value of the variable 'RESULTADO_FINAL' used by Base in line 15 of 'Config.java'. This creates a CF conflict that originates in lines 3 and 4 of 'Main.java' and goes from lines 7 and 11 of 'Config.java' to line 15 of 'Config.java'.

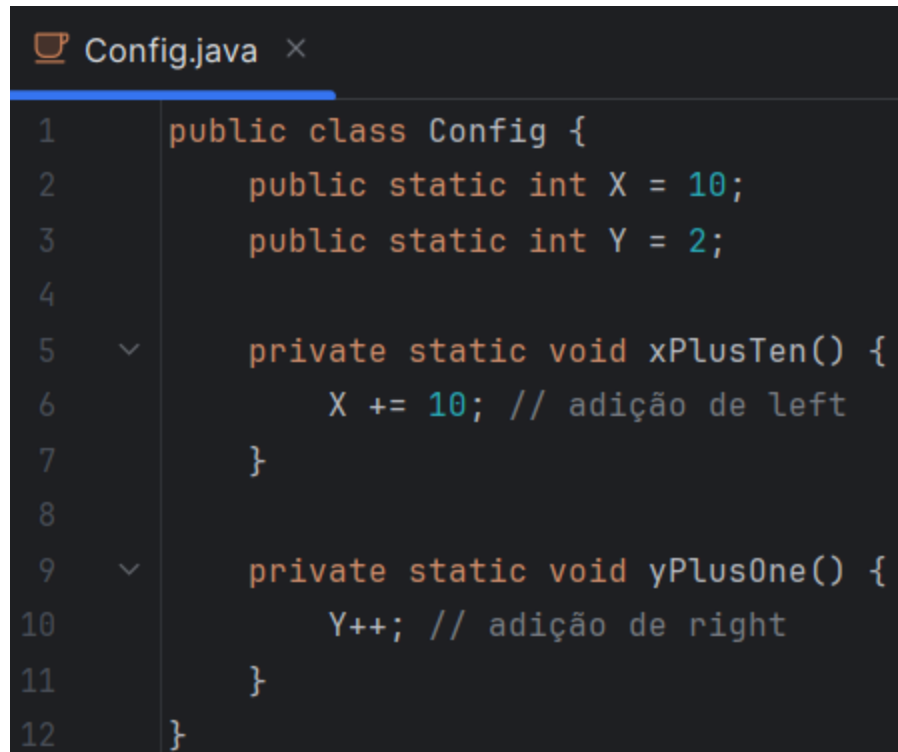
H2 - Left and Right add a method call in one file, which changes the value of different variables in a second file. In the first file, these two changes generate a CF conflict for a variable used by Base that uses both values changed by Left and Right..



The screenshot shows a code editor window titled 'Main.java'. The code is as follows:

```
1 public class Main { no usages
2     private static int RESULTADO_FINAL = 0; 2 usages
3
4     public static void main(String[] args) {
5         Config.xPlusTen(); // adição de left
6         Config.yPlusOne(); // adição de right
7         calcular();
8         System.out.println("RESULTADO_FINAL é " + RESULTADO_FINAL);
9     }
10
11     private static void calcular() { 1 usage
12         RESULTADO_FINAL = Config.X + Config.Y;
13     }
14 }
```

Line 3 is highlighted with a blue bar. Line 11 has a dropdown arrow next to it. The code shows two method calls in the 'main' method that modify variables in the 'Config' class, which then affect the 'RESULTADO_FINAL' variable used in the same 'main' method.



```
1 public class Config {
2     public static int X = 10;
3     public static int Y = 2;
4
5     private static void xPlusTen() {
6         X += 10; // adiç o de left
7     }
8
9     private static void yPlusOne() {
10        Y++; // adiç o de right
11    }
12 }
```

In line 5 of 'Main.java', Left added a method call that changes the value of 'X' in line 6 of 'Config.java'. In line 6 of 'Main.java', Right added a method call that changes the value of 'Y' in line 10 of 'Config.java'. These changes interfere with the value of the variable 'RESULTADO_FINAL' used by Base in line 12 of 'Main.java'. This creates a CF conflict originating in lines 5 and 6 of 'Main.java' and extending from lines 6 and 10 of 'Config.java' to line 12 of 'Main.java'.

I2 - Left adds a method call in file 1 that changes the value of a variable in file 2. Right adds a method call in file 2 that changes the value of another variable in file 1. At File 2, Base uses the values changed by Left and Right.

```

1 public class Main { no usages ykarocin *
2     public static int Y = 2; 1 usage
3
4     public static void main(String[] args) {
5         Config.xPlusTen(); // Left change
6         Config.processResult();
7     }
8
9     private static void yPlusOne(){ no usages
10        Y++; // Right change
11    }
12 }

```

```

1 public class Config { no usages
2     public static int X = 10; 3 usages
3     private static int RESULTADO_FINAL = 0; 2 usages
4
5     private static void xPlusTen(){ no usages
6         X += 10; // Left change
7     }
8
9     private static void processResult( no usages
10        Main.yPlusOne(); // Right change
11        RESULTADO_FINAL = X + Main.Y;
12        System.out.println(X + " + " + Main.Y + " = " + RESULTADO_FINAL);
13    )
14 }

```

Left adds a method call to line 5 of “Main.java” that changes the value of “X” on line 6 of “Config.java”. Right adds a method call to line 10 of “Config.java” that changes the value of “Y” on line 10 of “Main.java”. Base uses the values of the two variables changed by Left and Right on line 12 of “Config.java”. This generates a CF conflict that extends from lines 6 and 10 of “Main.java” to line 12 of “Config.java”.

A3 - Modifying a variable using Left in one file. Modifying a variable using Right in a second file. In a third file, there is a base variable that uses the values modified by Left and Right.

```

Config.java ×
1 public class Config { no usages
2     public static int X = 20; // alteração de left
3 }

```

```

Ajustador.java ×
1 public class Ajustador {
2     public static int Y = 3; // alteração de right
3 }

```

```

Main.java ×
1 public class Main {
2     private static int RESULTADO_FINAL = 0;
3
4     public static void main(String[] args) {
5         calcular();
6         System.out.println("RESULTADO_FINAL é " + RESULTADO_FINAL);
7     }
8
9     private static void calcular() {
10        RESULTADO_FINAL = Config.X + Ajustador.Y;
11    }
12 }

```

Left changes the value of the variable 'X' on line 2 of 'Config.java'. Right changes the value of the variable 'Y' on line 2 of 'Ajustador.java'. These changes interfere with the result of the variable 'RESULTADO_FINAL' used by Base on line 10 of 'Main.java'. This creates a CF conflict that extends from lines 2 of 'Config.java' and 2 of 'Ajustador.java' to line 10 of 'Main.java'.

B3- Left adds a method call in one file that modifies a variable in a second file. In a third file, another variable is modified by Right. Also in this third file, there is a base variable that uses the values modified by Left and Right.

```

Main.java ×
1 public class Main {
2     public static void main(String[] args) {
3         Config.xPlusTen(); // adição de left
4         Ajustador.calcular();
5         System.out.println("RESULTADO_FINAL é " + Ajustador.RESULTADO_FINAL);
6     }
7 }

```

```

Config.java ×
1 public class Config { no usages
2     public static int X = 10; 1 usage
3
4     private static void xPlusTen() { no usages
5         X += 10; // adição de left
6     }
7 }

```

```

Ajustador.java ×
1 public class Ajustador { no usages
2     public static int Y = 3; // alteração de right
3     private static int RESULTADO_FINAL = 0; 1 usage
4
5     private static void calcular() { no usages
6         RESULTADO_FINAL = Config.X + Y;
7     }
8 }

```

Left adds a method call to line 3 of 'Main.java' that changes the value of the variable 'X' on line 5 of 'Config.java'. Right changes the value of the variable 'Y' on line 2 of 'Ajustador.java'. These changes affect the result of the variable 'RESULTADO_FINAL' used by Base on line 6 of 'Ajustador.java'. This creates a CF conflict that originates on line 3 of 'Main.java' and goes from lines 5 of 'Config.java' and 2 of 'Ajustador.java' to line 6 of 'Ajustador.java'.

C3 - Left adds a method call in one file that modifies a variable in a second file. In a third file, another variable is modified by **Right**. In the second file, there is a base variable that uses the values modified by **Left** and **Right**.

Main.java ×

```
1 public class Main { no usages
2     public static void main(String[] args) {
3         Config.xPlusTen(); // adição de left
4         Config.calcular();
5         System.out.println("RESULTADO_FINAL é " + Config.RESULTADO_FINAL);
6     }
7 }
```

Config.java ×

```
1 public class Config {
2     public static int X = 10;
3     private static int RESULTADO_FINAL = 0;
4
5     private static void xPlusTen() {
6         X += 10; // adição de left
7     }
8
9     private static void calcular() {
10        RESULTADO_FINAL = X + Ajustador.Y;
11    }
12 }
```

Ajustador.java ×

```
1 public class Ajustador {
2     public static int Y = 3; // alteração de right
3 }
```

Left adds a method call to line 3 of 'Main.java' that changes the value of the variable 'X' on line 6 of 'Config.java'. Right changes the value of the variable 'Y' on line 2 of 'Ajustador.java'. These changes affect the result of the variable 'RESULTADO_FINAL' used by Base on line 10 of 'Config.java'. This creates a CF conflict originating on line 3 of 'Main.java' and going from lines 6 of 'Config.java' and 2 of 'Ajustador.java' to line 10 of 'Config.java'.

D3 - Left adds a method call in one file that modifies a variable in a second file. In the first file, another variable is modified by Right. In a third file, there is a base variable that uses the values modified by Left and Right.

```
Main.java ×
1 public class Main { no usages
2     public static int Y = 3; // alteração de right no usages
3
4     public static void main(String[] args) {
5         Config.xPlusTen(); // adição de left
6         Ajustador.calcular();
7         System.out.println("RESULTADO_FINAL é " + Ajustador.RESULTADO_FINAL);
8     }
9 }
```

```
Config.java ×
1 public class Config { no usages
2     public static int X = 10; 1 usage
3
4     private static void xPlusTen() {
5         X += 10; // adição de left
6     }
7 }
```

```

Ajustador.java ×
1  public class Ajustador {
2      private static int RESULTADO_FINAL = 0;
3
4      private static void calcular() {
5          RESULTADO_FINAL = Config.X + Main.Y;
6      }
7  }

```

Left adds a method call to line 5 of 'Main.java' that changes the value of the variable 'X' on line 5 of 'Config.java'. Right changes the value of the variable 'Y' on line 2 of 'Main.java'. These changes affect the result of the variable 'RESULTADO_FINAL' used by Base on line 5 of 'Ajustador.java'. This creates a CF conflict originating on line 5 of 'Main.java' and extending from lines 2 of 'Main.java' and 5 of 'Config.java' to line 5 of 'Ajustador.java'.

E3 - Left adds a method call in one file that modifies a variable in a second file. In a third file, another variable is modified by Right. In the first file, there is a base variable that uses the values modified by Left and Right.

```

Main.java ×
1  public class Main { no usages
2      private static int RESULTADO_FINAL = 0; 2 usages
3
4      public static void main(String[] args) {
5          Config.xPlusTen(); // adição de left
6          calcular();
7          System.out.println("RESULTADO_FINAL é " + RESULTADO_FINAL);
8      }
9
10     private static void calcular() { 1 usage
11         RESULTADO_FINAL = Config.X + Ajustador.Y;
12     }
13 }

```

```
Config.java ×  
1 public class Config {  
2     public static int X = 10;  
3  
4     private static void xPlusTen() {  
5         X += 10; // adição de left  
6     }  
7 }
```

```
Ajustador.java ×  
1 public class Ajustador { no usages  
2     public static int Y = 3; // alteração de right  
3 }
```

Left adds a method call to line 5 of 'Main.java' that changes the value of the variable 'X' on line 5 of 'Config.java'. Right changes the value of the variable 'Y' on line 2 of 'Ajustador.java'. These changes affect the result of the variable 'RESULTADO_FINAL' used by Base on line 11 of 'Main.java'. This creates a CF conflict originating on line 5 of 'Main.java' and extending from lines 5 of 'Config.java' and 2 of 'Ajustador.java' to line 11 of 'Main.java'.

F3 - Left adds a method call in one file that modifies a variable in a second file. In a third file, Right adds a method call that changes the value of another variable in the second file. Also in the second file, there is a base variable that uses the values modified by Left and Right.

Main.java ×

```
1 public class Main { no usages
2
3     public static void main(String[] args) {
4         Config.xPlusTen(); // adição de left
5         Ajustador.executar();
6         Config.calcular();
7         System.out.println("RESULTADO_FINAL é " + Config.RESULTADO_FINAL);
8     }
9 }
```

Config.java ×

```
1 public class Config { no usages
2     public static int X = 10; 2 usages
3     public static int Y = 2; 2 usages
4     private static int RESULTADO_FINAL = 0; 1
5
6     private static void xPlusTen() { no usages
7         X += 10; // adição de left
8     }
9
10    private static void yPlusOne() { no usages
11        Y++; // adição de right
12    }
13
14    private static void calcular() { no usages
15        RESULTADO_FINAL = X + Y;
16    }
17 }
```

```

1 public class Ajustador { no usages
2     private static void executar() { no usages
3         System.out.println("Executando Ajustador");
4         Config.yPlusOne(); // adição de right
5     }
6 }

```

Left adds a method call to line 4 of 'Main.java' that changes the value of the variable 'X' on line 7 of 'Config.java'. Right adds a method call to line 4 of 'Ajustador.java' that changes the value of the variable 'Y' on line 11 of 'Config.java'. These changes affect the result of the variable 'RESULTADO_FINAL' used by Base on line 15 of 'Config.java'. This creates a CF conflict originating from lines 4 of 'Main.java' and 4 of 'Ajustador.java', extending from lines 7 and 11 of 'Config.java' to line 15 of 'Config.java'.

G3 - Left adds a method call in one file that modifies a variable in a second file. In the first file, Right adds a method call that modifies the value of another variable in the second file. In a third file, there is a base variable that uses the values modified by Left and Right.

```

1 public class Main { no usages
2
3     public static void main(String[] args) {
4         Config.xPlusTen(); // adição de left
5         Config.yPlusOne(); // adição de right
6         Ajustador.calcular();
7         System.out.println("RESULTADO_FINAL é " + Ajustador.RESULTADO_FINAL);
8     }
9 }

```

```

Config.java ×
1  public class Config {
2      public static int X = 10;
3      public static int Y = 2;
4
5      private static void xPlusTen() {
6          X += 10; // adição de left
7      }
8
9      private static void yPlusOne() {
10         Y++; // adição de right
11     }
12 }

```

```

Ajustador.java ×
1  public class Ajustador {
2      private static int RESULTADO_FINAL = 0;
3
4      private static void calcular() {
5          RESULTADO_FINAL = Config.X + Config.Y;
6      }
7  }

```

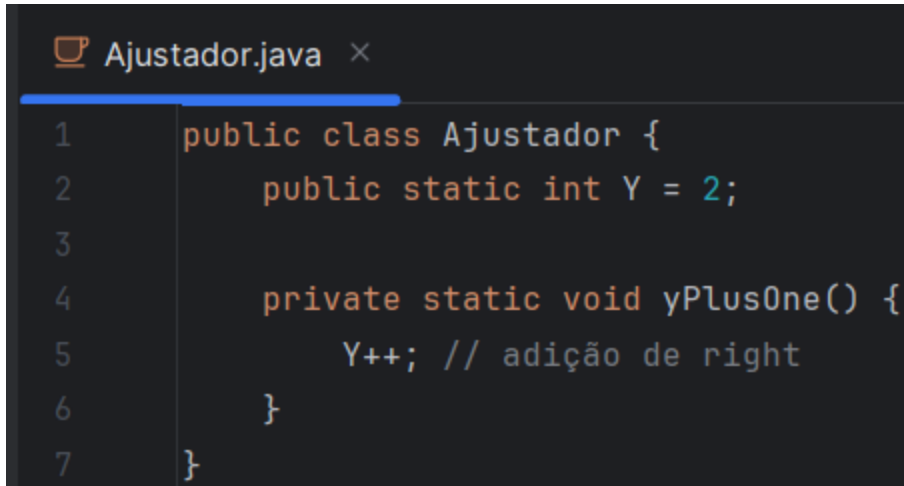
Left adds a method call to line 4 of 'Main.java' that changes the value of the variable 'X' on line 6 of 'Config.java'. Right adds a method call to line 5 of 'Main.java' that changes the value of the variable 'Y' on line 10 of 'Config.java'. These changes affect the result of the variable 'RESULTADO_FINAL' used by Base on line 5 of 'Ajustador.java'. This creates a CF conflict originating in lines 4 and 5 of 'Main.java' and extending from lines 6 and 10 of 'Config.java' to line 5 of 'Ajustador.java'.

H3 - Left adds a method call in one file that modifies a variable in a second file. In the first file, Right adds a method call that modifies the value of another variable

in a third file. In the second file, there is a base variable that uses the values modified by Left and Right.

```
☞ Main.java ×  
1 public class Main { no usages  
2     public static void main(String[] args) {  
3         Config.xPlusTen(); // adição de left  
4         Ajustador.yPlusOne(); // adição de right  
5         Config.calcular();  
6         System.out.println("RESULTADO_FINAL é " + Config.RESULTADO_FINAL);  
7     }  
8 }
```

```
☞ Config.java ×  
1 public class Config { 1 usage  
2     public static int X = 10; 2 usages  
3     private static int RESULTADO_FINAL = 0;  
4  
5     private static void xPlusTen() { no usage  
6         X += 10; // adição de left  
7     }  
8  
9     private static void calcular() { no usage  
10        RESULTADO_FINAL = X + Ajustador.Y;  
11    }  
12 }
```



```
Ajustador.java x
1 public class Ajustador {
2     public static int Y = 2;
3
4     private static void yPlusOne() {
5         Y++; // adição de right
6     }
7 }
```

Left adds a method call to line 3 of 'Main.java' that changes the value of the variable 'X' on line 6 of 'Config.java'. Right adds a method call to line 4 of 'Main.java' that changes the value of the variable 'Y' on line 5 of 'Ajustador.java'. These changes affect the result of the variable 'RESULTADO_FINAL' used by Base on line 10 of 'Config.java'. This creates a CF conflict originating from lines 3 and 4 of 'Main.java' and extending from lines 6 of 'Config.java' and 5 of 'Ajustador.java' to line 10 of 'Config.java'.

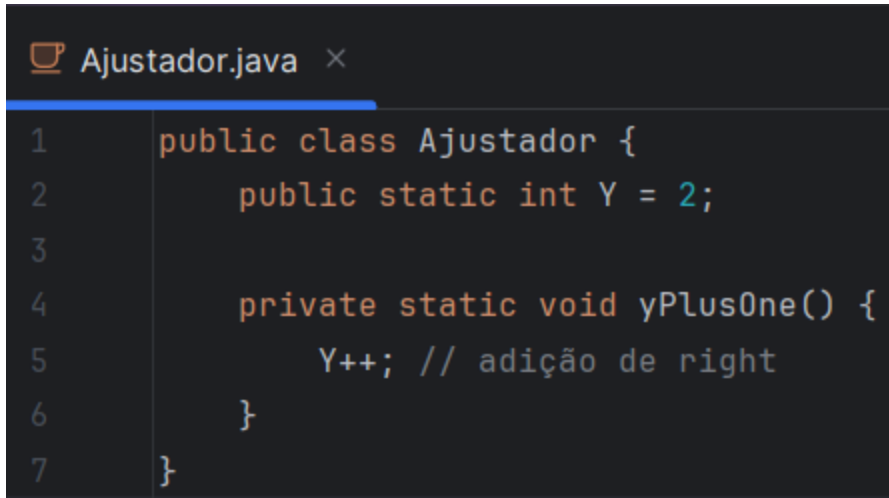
I3 - Left adds a method call in one file that modifies a variable in a second file. In the first file, Right adds a method call that modifies the value of another variable in a third file. In the first file, there is a base variable that uses the values modified by Left and Right.

Main.java ×

```
1 public class Main { no usages
2     private static int RESULTADO_FINAL = 0; 2 usages
3
4     public static void main(String[] args) {
5         Config.xPlusTen(); // adição de left
6         Ajustador.yPlusOne(); // adição de right
7         calcular();
8         System.out.println("RESULTADO_FINAL é " + RESULTADO_FINAL);
9     }
10
11     private static void calcular() { 1 usage
12         RESULTADO_FINAL = Config.X + Ajustador.Y;
13     }
14 }
```

Config.java ×

```
1 public class Config { no usages
2     public static int X = 10; 1 usage
3
4     private static void xPlusTen() {
5         X += 10; // adição de left
6     }
7 }
```



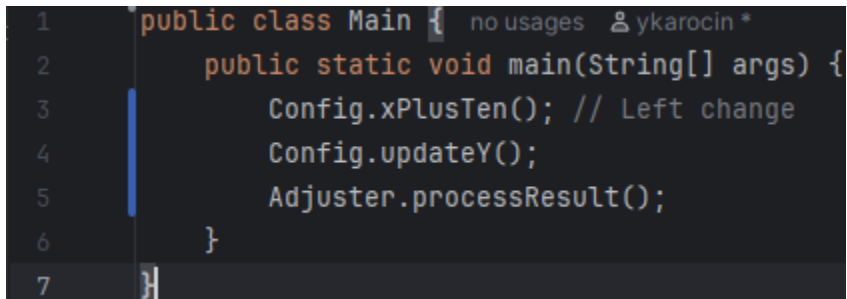
```

1 public class Ajustador {
2     public static int Y = 2;
3
4     private static void yPlusOne() {
5         Y++; // adição de right
6     }
7 }

```

Left adds a method call to line 5 of 'Main.java' that changes the value of the variable 'X' in line 5 of 'Config.java'. Right adds a method call to line 6 of 'Main.java' that changes the value of the variable 'Y' in line 5 of 'Ajustador.java'. These changes interfere with the result of the variable 'RESULTADO_FINAL' used by Base in line 12 of 'Main.java'. This generates a CF conflict originating in lines 5 and 6 of 'Main.java' and extending from lines 5 of 'Config.java' and 5 of 'Ajustador.java' to line 12 of 'Main.java'.

J3 - Left adds a method call in file 1 that changes the value of a variable in file 2. Right changes the value of another variable in file 2. In file 3 there is a base variable that uses the values changed by Left and Right.



```

1 public class Main { no usages  ykarocin *
2     public static void main(String[] args) {
3         Config.xPlusTen(); // Left change
4         Config.updateY();
5         Adjuster.processResult();
6     }
7 }

```

```

1 public class Config { no usages
2     public static int X = 10; 1 usage
3     public static int Y = 2; 1 usage
4
5     private static void xPlusTen(){
6         X += 10; // Left change
7     }
8
9     private static void updateY(){
10        Y += 2; // Right change
11    }
12 }

```

```

1 public class Adjuster { no usages new *
2     private static int RESULTADO_FINAL = 0; 2 usages
3
4     private static void processResult() { no usages new *
5         RESULTADO_FINAL = Config.X + Config.Y;
6         System.out.println(Config.X + " + " + Config.Y + " = " + RESULTADO_FINAL);
7     }
8 }

```

In line 3 of “Main.java”, Left adds a method call that changes the value of “X” in line 6 of “Config.java”. Right changes the value of “Y” in line 10 of “Config.java”. In line 5 of “Adjuster.java”, Base uses both values in the variable “RESULTADO_FINAL”. This generates a CF conflict between lines 6 and 10 of “Config.java” and line 5 of “Adjuster.java”.

K3 - Left adds a method call in file 1 that changes the value of a variable in file 2. Right adds a method call in file 2 that changes the value of another variable in file 1. In a third file, there is a base variable that uses the values changed by Left and Right.

```

1 public class Main { no usages ykarocin *
2     public static int Y = 2; 1 usage
3
4     public static void main(String[] args) {
5         Config.xPlusTen(); // Left change
6         Config.processResult();
7     }
8
9     private static void updateY(){ no usages
10         Y += 2; // Right change
11     }
12 }

```

```

1 public class Config { no usages
2     public static int X = 10; 1 usage
3
4     private static void xPlusTen(){ no usages
5         X += 10; // Left change
6     }
7
8     private static void processResult(){
9         Main.updateY(); // Right change
10        Adjuster.sumAll();
11    }
12 }

```

```

1 public class Adjuster { no usages new *
2     private static int RESULTADO_FINAL = 0; 2 usages
3
4     private static void sumAll() { no usages new *
5         RESULTADO_FINAL = Config.X + Main.Y;
6         System.out.println(Config.X + " + " + Main.Y + " = " + RESULTADO_FINAL);
7     }
8 }

```

On line 5 of “Main.java”, Left adds a method call that changes the value of “X” on line 5 of “Config.java”. On line 9 of “Config.java”, Right adds a method call that changes the value of “Y” on line 10 of “Main.java”. Base uses the values of “X” and “Y” in the variable “RESULTADO_FINAL” on line 5 of “Adjuster.java”. This generates a CF conflict between lines 5 of “Config.java” and 10 of “Main.java” and line 5 of “Adjuster.java”.

L3 - Left adds a method call in file 1 that changes the value of a variable in file 2. Right adds a method call in file 2 that changes the value of a variable in file 3. In file 1 there is a base variable that uses the values changed by Left and Right.

```
1 public class Main {  @ ykarocin *
2     private static int RESULTADO_FINAL = 0;
3
4     public static void main(String[] args) {  @ ykarocin *
5         Config.xPlusTen(); // Left change
6         Config.processResult();
7     }
8
9     private static void sumAll() { new *
10        RESULTADO_FINAL = Config.X + Adjuster.Y;
11        System.out.println(Config.X + " + " + Adjuster.Y + " = " + RESULTADO_FINAL);
12    }
13 }
```

```
1 public class Config { no usages
2     public static int X = 10; 1 usage
3
4     private static void xPlusTen(){ no usages
5         X += 10; // Left change
6     }
7
8     private static void processResult(){ no usages
9         Adjuster.updateY(); // Right change
10        Main.sumAll();
11    }
12 }
```

```
1 public class Adjuster { no usages ne
2     public static int Y = 2; 1 usage
3
4     private static void updateY(){
5         Y += 2; // Right change
6     }
7 }
```

In line 5 of “Main.java”, Left adds a method call that changes the value of “X” in line 5 of “Config.java”. In line 9 of “Config.java”, Right adds a method call that changes the value of “Y” in line 5 of “Adjuster.java”. Base uses the values of “X” and “Y” in the variable

“RESULTADO_FINAL” in line 10 of “Main.java”. This generates a CF conflict between lines 5 of “Config.java” and 5 of “Adjuster.java” and line 10 of “Main.java”.

M3 - Left adds a method call in file 1 that changes the value of a variable in file 2. Right adds a method call in file 2 that changes the value of a variable in file 3. In file 2 there is a base variable that uses the values changed by Left and Right.

```
1 public class Main { no usages 3 ykarocin *
2     public static void main(String[] args) {
3         Config.xPlusTen(); // Left change
4         Config.processResult();
5     }
6 }
```

```
1 public class Config { no usages
2     public static int X = 10; 3 usages
3     private static int RESULTADO_FINAL = 0; 2 usages
4
5     private static void xPlusTen(){ no usages
6         X += 10; // Left change
7     }
8
9     private static void processResult(){ no usages
10        Adjuster.updateY(); // Right change
11        sumAll();
12    }
13
14    private static void sumAll() { 1 usage
15        RESULTADO_FINAL = X + Adjuster.Y;
16        System.out.println(X + " + " + Adjuster.Y + " = " + RESULTADO_FINAL);
17    }
18 }
```

```
1 public class Adjuster { no usages ne
2     public static int Y = 2; 1 usage
3
4     private static void updateY(){
5         Y += 2; // Right change
6     }
7 }
```

In line 3 of “Main.java”, Left adds a method call that changes the value of “X” in line 6 of “Config.java”. In line 10 of “Config.java”, Right adds a method call that changes the value of “Y” in line 5 of “Adjuster.java”. Base uses the values of “X” and “Y” in the variable “RESULTADO_FINAL” in line 15 of “Config.java”. This generates a CF conflict between lines 6 of “Config.java” and 5 of “Adjuster.java” and line 15 of “Main.java”.

N3 - Left adds a method call in file 1 that changes the value of a variable in file 2. Right adds a method call in file 2 that changes the value of a variable in file 3. In file 3 there is a base variable that uses the values changed by Left and Right.

```
1  public class Main { no usages 3 ykarocin *
2      public static void main(String[] args) {
3          Config.xPlusTen(); // Left change
4          Config.processResult();
5      }
6  }
```

```
1  public class Config { no usages
2      public static int X = 10; 1 usage
3
4      private static void xPlusTen(){ no usages
5          X += 10; // Left change
6      }
7
8      private static void processResult(){ no
9          Adjuster.updateY(); // Right change
10         Adjuster.sumAll();
11     }
12 }
```

```

1  public class Adjuster { no usages new *
2      public static int Y = 2; 3 usages
3      private static int RESULTADO_FINAL = 0; 2 usages
4
5      private static void updateY(){ no usages new *
6          Y += 2; // Right change
7      }
8
9      private static void sumAll() { no usages new *
10         RESULTADO_FINAL = Config.X + Y;
11         System.out.println(Config.X + " + " + Y + " = " + RESULTADO_FINAL);
12     }
13 }

```

In line 3 of “Main.java”, Left adds a method call that changes the value of “X” in line 5 of “Config.java”. In line 9 of “Config.java”, Right adds a method call that changes the value of “Y” in line 6 of “Adjuster.java”. Base uses the values of “X” and “Y” in the variable “RESULTADO_FINAL” in line 10 of “Adjuster.java”. This generates a CFconflict between lines 5 of “Config.java” and 6 of “Adjuster.java” and line 10 of “Adjuster.java”.

O3 - Left adds a method call in file 1 that changes the value of a variable in file 2. Right adds a method call in file 3 that changes the value of a variable in file 2. In file 1 there is a base variable that uses the values changed by Left and Right.

```

1  public class Main { no usages 3 ykarocin *
2      private static int RESULTADO_FINAL = 0; 2 usages
3
4      public static void main(String[] args) { no usages 3 ykarocin *
5          Config.xPlusTen(); // Left change
6          Adjuster.processResult();
7      }
8
9      private static void sumAll() { no usages new *
10         RESULTADO_FINAL = Config.X + Config.Y;
11         System.out.println(Config.X + " + " + Config.Y + " = " + RESULTADO_FINAL);
12     }
13 }

```

```

1      public class Config { no usages
2          public static int X = 10; 1 usage
3          public static int Y = 2; 1 usage
4
5          private static void xPlusTen(){
6              X += 10; // Left change
7          }
8
9          private static void updateY(){
10             Y += 2; // Right change
11         }
12     }

```

```

1      public class Adjuster { no usages new *
2          private static void processResult(){
3              Config.updateY(); // Right change
4              Main.sumAll();
5          }
6      }

```

In line 5 of “Main.java”, Left adds a method call that changes the value of “X” in line 6 of “Config.java”. In line 3 of “Adjuster.java”, Right adds a method call that changes the value of “Y” in line 10 of “Config.java”. Base uses the values of “X” and “Y” in the variable “RESULTADO_FINAL” in line 10 of “Main.java”. This generates a CFconflict between lines 6 and 10 of “Config.java” and line 10 of “Main.java”.

A4 - Left adds a method call in one file that modifies a variable in a second file. In a third file, Right modifies the value of another variable. In a fourth file, there is a base variable that uses the values modified by Left and Right.

```

Main.java x
1      public class Main { no usages
2          public static void main(String[] args) {
3              Config.xPlusTen(); // adição de left
4              Adjustador.checkY();
5              Calculador.calcular();
6              System.out.println("RESULTADO_FINAL é " + Calculador.RESULTADO_FINAL);
7          }
8      }

```

☕ Config.java ×

```
1 public class Config { no usages
2     public static int X = 10; 1 usage
3
4     private static void xPlusTen() {
5         X += 10; // adição de left
6     }
7 }
```

☕ Ajustador.java ×

```
1 public class Ajustador { no usages
2     public static int Y = 2; 1 usage
3
4     private static void checkY() {
5         Y++; // adição de right
6     }
7 }
```

☕ Calculador.java ×

```
1 public class Calculador { no usages
2     private static int RESULTADO_FINAL = 0; 1 usage
3
4     private static void calcular() { no usages
5         RESULTADO_FINAL = Config.X + Ajustador.Y;
6     }
7 }
```

Left adds a method call to line 3 of 'Main.java' that changes the value of the variable 'X' on line 5 of 'Config.java'. Right changes the value of the variable 'Y' on line 5 of 'Ajustador.java'. These changes affect the result of the variable 'RESULTADO_FINAL' used by Base on line 5 of 'Calculador.java'. This creates a CFconflict originating on line

3 of 'Main.java' and extending from lines 5 of 'Config.java' and 5 of 'Ajustador.java' to line 5 of 'Calculador.java'.

B4 - Left adds a method call in one file that modifies a variable in a second file. In a third file, Right adds a method call that modifies another variable in a fourth file. In the second file, there is a base variable that uses the values modified by Left and Right.

```
☞ Main.java ×  
1 public class Main { no usages  
2     public static void main(String[] args) {  
3         Config.xPlusTen(); // adição de left  
4         Ajustador.checkY();  
5         Config.calcular();  
6         System.out.println("RESULTADO_FINAL é " + Config.RESULTADO_FINAL);  
7     }  
8 }
```

```
☞ Config.java ×  
1 public class Config { no usages  
2     public static int X = 10; 2 usages  
3     private static int RESULTADO_FINAL = 0;  
4  
5     private static void xPlusTen() { no usages  
6         X += 10; // adição de left  
7     }  
8  
9     private static void calcular() { no usages  
10        RESULTADO_FINAL = X + Ajustador.Y;  
11    }  
12 }
```

```

Ajustador.java x
1  public class Ajustador {
2      public static int Y = 2;
3
4      private static void checkY() {
5          if ( Y > 0 ) {
6              Calculador.plusOneY(); // adição de right
7          } else {
8              Y = 1;
9              Calculador.plusOneY();
10         }
11     }
12 }

```

```

Calculador.java x
1  public class Calculador { no usages
2      private static void plusOneY() { no us
3          Ajustador.Y++; // adição de right
4      }
5  }

```

Left adds a method call to line 3 of 'Main.java' that changes the value of the variable 'X' on line 6 of 'Config.java'. Right adds a method call to line 6 of 'Ajustador.java' that changes the value of the variable 'Y' on line 3 of 'Calculador.java'. These changes affect the result of the variable 'RESULTADO_FINAL' used by Base on line 10 of 'Config.java'. This creates a CF conflict originating from lines 3 of 'Main.java' and 6 of 'Ajustador.java', extending from lines 6 of 'Config.java' and 3 of 'Calculador.java' to line 10 of 'Config.java'.

C4 - Left adds a method call in one file that modifies a variable in a second file. In a third file, Right adds a method call that modifies another variable in a fourth file. In the first file, there is a base variable that uses the values modified by Left and Right.

☞ Main.java ×

```
1  public class Main { no usages
2      private static int RESULTADO_FINAL = 0;
3
4      public static void main(String[] args) {
5          Config.xPlusTen(); // adição de left
6          Ajustador.checkY();
7          calcular();
8          System.out.println("RESULTADO_FINAL é " + RESULTADO_FINAL);
9      }
10
11     private static void calcular() {
12         RESULTADO_FINAL = Config.X + Ajustador.Y;
13     }
14 }
```

☞ Config.java ×

```
1  public class Config {
2      public static int X = 10;
3
4      private static void xPlusTen() {
5          X += 10; // adição de left
6      }
7  }
```

```
Ajustador.java ×
1 public class Ajustador {
2     public static int Y = 2;
3
4     private static void checkY() {
5         if ( Y > 0 ) {
6             Calculador.plusOneY(); // adição de right
7         } else {
8             Y = 1;
9             Calculador.plusOneY();
10        }
11    }
12 }
```

```
Calculador.java ×
1 public class Calculador { no usages
2     private static void plusOneY() { no us
3         Ajustador.Y++; // adição de right
4     }
5 }
```

Left adds a method call to line 5 of 'Main.java' that changes the value of the variable 'X' on line 5 of 'Config.java'. Right adds a method call to line 6 of 'Ajustador.java' that changes the value of the variable 'Y' on line 3 of 'Calculador.java'. These changes affect the result of the variable 'RESULTADO_FINAL' used by Base on line 12 of 'Main.java'. This creates a CF conflict originating from lines 5 of 'Main.java' and 6 of 'Ajustador.java', extending from lines 5 of 'Config.java' and 3 of 'Calculador.java' to line 12 of 'Main.java'.

D4 - Left adds a method call in one file that modifies a variable in a second file. In a third file, Right adds a method call that modifies another variable, also in the second file. In a fourth file, there is a base variable that uses the values modified by Left and Right.

☞ Main.java ×

```
1 public class Main { no usages
2     public static void main(String[] args) {
3         Config.xPlusTen(); // adição de left
4         Ajustador.checkY();
5         Calculador.calcular();
6         System.out.println("RESULTADO_FINAL é " + Calculador.RESULTADO_FINAL);
7     }
8 }
```

☞ Config.java ×

```
1 public class Config { no usages
2     public static int X = 10; 1 usage
3
4     private static void xPlusTen() { no us
5         X += 10; // adição de left
6     }
7
8     private static void plusOneY() { no us
9         Ajustador.Y++; // adição de right
10    }
11 }
```

```

Ajustador.java ×
1  public class Ajustador {
2      public static int Y = 2;
3
4      private static void checkY() {
5          if ( Y > 0 ) {
6              Config.plusOneY(); // adição de right
7          } else {
8              Y = 1;
9              Config.plusOneY();
10         }
11     }
12 }

```

```

Calculador.java ×
1  public class Calculador { no usages
2      private static int RESULTADO_FINAL = 0; 1 usage
3
4      private static void calcular() { no usages
5          RESULTADO_FINAL = Config.X + Ajustador.Y;
6      }
7  }

```

Left adds a method call to line 3 of 'Main.java' that changes the value of the variable 'X' on line 5 of 'Config.java'. Right adds a method call to line 6 of 'Ajustador.java' that changes the value of the variable 'Y' on line 9 of 'Config.java'. These changes affect the result of the variable 'RESULTADO_FINAL' used by Base on line 5 of 'Calculador.java'. This creates a CF conflict originating from lines 3 of 'Main.java' and 6 of 'Ajustador.java', extending from lines 5 and 9 of 'Config.java' to line 5 of 'Calculador.java'.

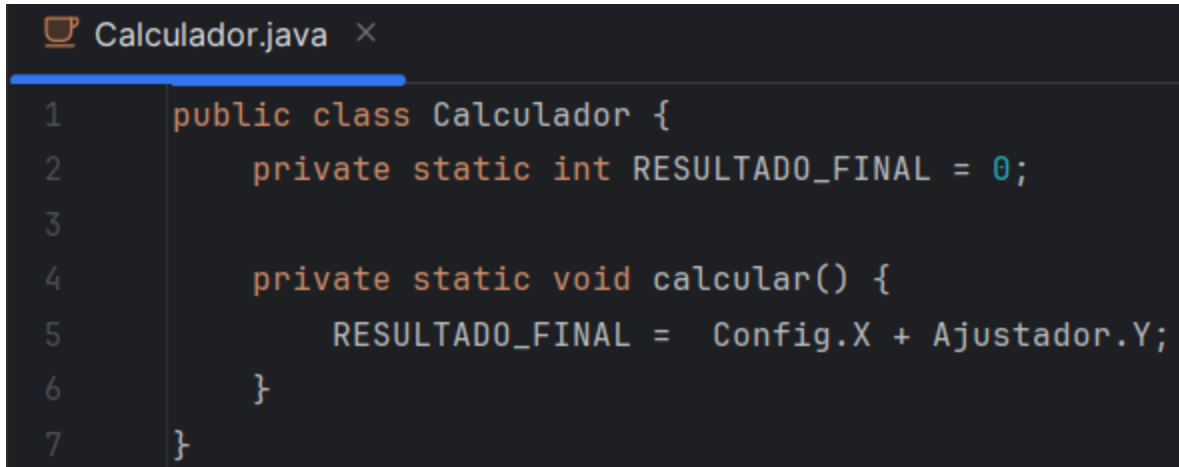
E4 - Left adds a method call in one file that modifies a variable in a second file. Also in the first file, Right adds a method call that modifies another variable in a

third file. In a fourth file, there is a base variable that uses the values modified by Left and Right.

```
☞ Main.java ×  
1 public class Main { no usages  
2     public static void main(String[] args) {  
3         Config.xPlusTen(); // adição de left  
4         Ajustador.yPlusOne(); // adição de right  
5         Calculador.calcular();  
6         System.out.println("RESULTADO_FINAL é " + Calculador.RESULTADO_FINAL);  
7     }  
8 }
```

```
☞ Config.java ×  
1 public class Config { no usages  
2     public static int X = 10; 1 usage  
3  
4     private static void xPlusTen()  
5         X += 10; // adição de left  
6     }  
7 }
```

```
☞ Ajustador.java ×  
1 public class Ajustador { no usages  
2     public static int Y = 2; 1 usage  
3  
4     private static void yPlusOne() {  
5         Y++; // adição de right  
6     }  
7 }
```



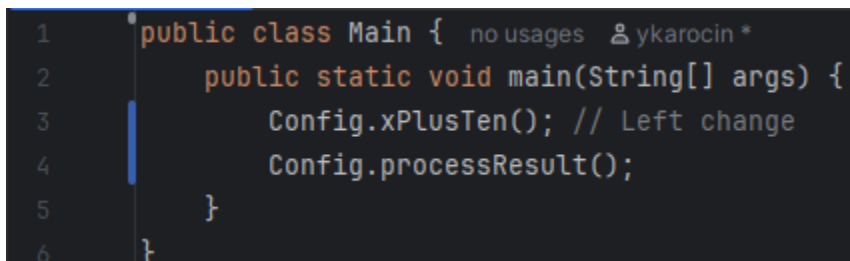
```

Calculador.java x
1  public class Calculador {
2      private static int RESULTADO_FINAL = 0;
3
4      private static void calcular() {
5          RESULTADO_FINAL = Config.X + Ajustador.Y;
6      }
7  }

```

Left adds a method call to line 3 of 'Main.java' that changes the value of the variable 'X' on line 5 of 'Config.java'. Right adds a method call to line 4 of 'Main.java' that changes the value of the variable 'Y' on line 5 of 'Ajustador.java'. These changes affect the result of the variable 'RESULTADO_FINAL' used by Base on line 5 of 'Calculador.java'. This creates a CF conflict originating from lines 3 and 4 of 'Main.java' and extending from lines 5 of 'Config.java' and 5 of 'Ajustador.java' to line 5 of 'Calculador.java'.

F4 - Left adds a method call in file 1 that changes the value of a variable in file 2. Right adds a method call in file 2 that changes the value of a variable in file 3. In file 4 there is a base variable that uses the values changed by Left and Right.



```

1  public class Main { no usages  3 ykarocin *
2      public static void main(String[] args) {
3          Config.xPlusTen(); // Left change
4          Config.processResult();
5      }
6  }

```

```

1 public class Config { no usages
2     public static int X = 10; 1 usage
3
4     private static void xPlusTen(){ no usages
5         X += 10; // Left change
6     }
7
8     private static void processResult(){ no
9         Adjuster.updateY(); // Right change
10        Calculator.sumAll();
11    }
12 }

```

```

1 public class Adjuster { no usages ne
2     public static int Y = 2; 1 usage
3
4     private static void updateY(){
5         Y += 2; // Right change
6     }
7 }

```

```

1 public class Calculator { no usages new *
2     private static int RESULTADO_FINAL = 0; 2 usages
3
4     private static void sumAll() { no usages new *
5         RESULTADO_FINAL = Config.X + Adjuster.Y;
6         System.out.println(Config.X + " + " + Adjuster.Y + " = " + RESULTADO_FINAL);
7     }
8 }

```

In line 3 of “Main.java”, Left adds a method call that changes the value of “X” in line 5 of “Config.java”. In line 9 of “Config.java”, Right adds a method call that changes the value of “Y” in line 5 of “Adjuster.java”. Base uses the values of “X” and “Y” in the variable “RESULTADO_FINAL” in line 5 of “Calculator.java”. This generates a CF conflict between lines 6 of “Config.java” and 5 of “Adjuster.java” and line 5 of “Calculator.java”.

A5 - Left adds a method call in one file that modifies a variable in a second file. In a third file, Right adds a method call that modifies another variable in a fourth file. In a fifth file, there is a base variable that uses the values modified by Left and Right.

☞ Main.java ×

```
1 public class Main { no usages
2     public static void main(String[] args) {
3         Config.xPlusTen(); // adição de left
4         Ajustador.checkY();
5         Autenticador.calcular();
6         System.out.println("RESULTADO_FINAL é " + Autenticador.RESULTADO_FINAL);
7     }
8 }
```

☞ Config.java ×

```
1 public class Config {
2     public static int X = 10;
3
4     private static void xPlusTen() {
5         X += 10; // adição de left
6     }
7 }
```

☞ Ajustador.java ×

```
1 public class Ajustador { no usages
2     public static int Y = 2; 2 usage
3
4     private static void checkY() {
5         // adição de right
6         if (Y > 0) {
7             Calculador.yPlusOne();
8         } else {
9             Y = 1;
10            Calculador.yPlusOne();
11        }
12    }
```

```
☞ Calculador.java ×  
1 public class Calculador { no usages  
2     private static void yPlusOne() { no us  
3         Ajustador.Y++; // adição de right  
4     }  
5 }
```

```
☞ Autenticador.java ×  
1 public class Autenticador {  
2     private static int RESULTADO_FINAL = 0;  
3  
4     private static void calcular() {  
5         RESULTADO_FINAL = Config.X + Ajustador.Y;  
6     }  
7 }
```

Left adds a method call to line 3 of 'Main.java' that changes the value of the variable 'X' on line 5 of 'Config.java'. Right adds a method call to line 7 of 'Ajustador.java' that changes the value of the variable 'Y' on line 3 of 'Calculador.java'. These changes affect the result of the variable 'RESULTADO_FINAL' used by Base on line 5 of 'Autenticador.java'. This creates a CF conflict originating from lines 3 of 'Main.java' and 7 of 'Ajustador.java', and extending from lines 5 of 'Config.java' and 3 of 'Calculador.java' to line 5 of 'Autenticador.java'.